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Issuance Overpricing of China's Corporate Debt Securities

Contents

1 Big picture

- **Question.** Why are new issues of Chinese corporate debt *overpriced* relative to their first secondary-market price, and what institutional channels generate this result?
- **Setting.** China's interbank corporate debt market, 2015–2019. Banks are the dominant underwriters and investors, debt maturities are short, and issues are sold through single-price auctions.
- **Main fact.** The average first-trade spread is **4.9 bps** above the issuance spread, and the first-trade excess return is **-7.67 bps**. Both indicate issuance overpricing.
- **Policy shock.** After the October 2017 ban on underwriter rebates, average overpricing falls from **7.44 bps** to **2.41 bps**, but remains positive.
- **Mechanism.** Because firms issue repeatedly and issuance yields are public financing benchmarks, issuers reward underwriters for higher issuance prices. Underwriters respond by using rebates and, later, self-purchases/overbidding.
- **Contribution.** The paper explains why China's primary debt market displays the opposite sign from the underpricing commonly seen in Western markets.

2 Data and measurement (key objects)

2.1 Institutional setting

- The paper studies non-financial corporate debt in China's *interbank* market, which accounts for about 90% of issuance.
- The market is over-the-counter and populated by qualified institutions such as banks, funds, insurers, and securities firms.
- The relevant securities are mainly CP and MTN; together they account for more than 86% of non-financial issuance in the market.
- Issues are sold in a **single-price auction**. Underwriters do not allocate the issue discretionarily, but they can bid for their own account or for clients.
- Underwriters are not obliged to stabilize secondary-market prices after issuance.

2.2 Sample and sources

- **Security sample.** All CP and MTN issued by non-financial firms from 2015 to 2019.
- **Sources.** WIND and CFETS for issuance and issuer characteristics; WIND and Choice for secondary-market prices; confidential NAFMII auction-allocation data for underwriter purchases.
- **Initial sample.** 19,510 debt securities with total issuance over 23 trillion RMB.
- **Main sample.** After requiring at least one trade within seven calendar days of the first trading day, the sample contains **18,229** issues by **2,558** firms. Of these, **17,709** trade on the first trading day.
- **Auction-allocation sample.** 17,373 issues with institution-level purchase data.

2.3 Important descriptive facts

- Mean maturity is **1.74 years**; mean issue size is **1.156 billion RMB**.
- Mean subscription ratio is **1.74**; median is **1.49**.
- Only about **7%** of issues are first-time issuances; about **84%** come from firms that issued in the previous year.
- Ratings are highly skewed upward; the median issue is **AA+**.
- Issuers are large firms: mean assets are about **164 billion RMB**.
- Underwriters acquire on average **35%** of the issues they underwrite.

2.4 Overpricing measures

- **Primary measure.** Spread change from issuance to first trade, relative to a Treasury yield of similar maturity.
- **Alternative measure.** Excess return from issuance to first trade, benchmarked to a same-rating CSI corporate debt index.
- **Longer-horizon check.** The authors also compute overpricing 15 calendar days after issuance.
- **Secondary-market liquidity benchmark.** Estimated first-month trading cost is about **10.11 bps**, which helps explain why investors may still prefer the primary market even when issues are overpriced.

3 Models and empirical specifications (all equations + interpretation)

3.1 Main overpricing measure

$$\Delta Spread_i = Spread_{i,\text{first trade}} - Spread_{i,\text{issuance}}. \tag{1}$$

- **Interpretation.** A positive $\Delta Spread_i$ means the issue's yield spread rises once it starts trading, so the issuance price was too high.
- **Economic meaning.** Because yield is the issuer's financing cost, a positive spread change means the issuer borrowed more cheaply at issuance than the market subsequently supports.

3.2 Excess-return measure

$$Ret_i = \frac{P_{i,T} - P_{i,t}}{P_{i,t}}, \tag{2}$$

where $P_{i,t}$ is the issuance price and $P_{i,T}$ is the first-trade price including accrued interest.

- **Interpretation.** A negative benchmark-adjusted first-trade return means investors who bought at issuance lost money relative to comparable bonds.
- **Relation to the first measure.** The spread-change and excess-return measures should line up because price and yield move in opposite directions.

3.3 Underwriter-retention specification (implicit in Table 4)

A convenient way to summarize Table 4 is

$$\Pr\left(Switch_{j,n+1} = 1\right) = \Lambda(\alpha + \beta\, Underperformed_{j,n} + \gamma' X_{j,n})\,, \tag{3}$$

where $Underperformed_{j,n} = 1$ if issue n's issuance spread is above the benchmark spread of comparable issues, and $X_{j,n}$ contains issuance and issuer controls.

- **Interpretation.** If $\beta > 0$, pricing poorly today raises the probability that the issuer switches underwriters next time.
- **Why it matters.** This is the paper's direct evidence that underwriters have a repeat-business incentive to push issuance prices up.

3.4 Difference-in-differences around the rebate ban

$$\Delta Spread_{i,j,t} = \theta_0 + \theta_1 Treat_j + \theta_2 Post_t + \theta_3(Treat_j \times Post_t) + \sum_m \theta_m Control_{m,i,j} + \varepsilon_{i,j,t}. \tag{4}$$

- **Treatments.** In one set of regressions, $Treat_j = 1$ for central SOE issuers. In another, $Treat_j = 1$ for Big Four underwriters.
- **Post.** $Post_t = 1$ after October 1, 2017.
- **Interpretation of θ_3.** The interaction coefficient is the DiD estimate of how the rebate ban changed overpricing differentially in the treatment group.
- **Prediction.** If rebates were most valuable where competition for mandates was strongest, central-SOE issues should show a larger post-ban drop, while Big Four underwriters should show a smaller drop.

3.5 Overbidding/self-purchase specification

$$\Delta Spread_{i,j} = \theta_0 + \theta_1 UnderwriterShare_i + \sum_m \theta_m Control_{m,i,j} + \varepsilon_{i,j}. \tag{5}$$

- **Interpretation.** $UnderwriterShare_i$ is the fraction of the issue bought by its underwriter(s).
- **Competing hypotheses.** If underwriters buy because issues are undervalued or because they provide fair-value support, θ_1 should not be strongly positive. If they overbid to win future business, θ_1 should be positive.

3.6 Quality-of-pricing regression (implicit in Table 9)

$$Spread^{issuance}_{i,j,t} = \alpha + X'_{i,j,t}\beta + u_{i,j,t}. \tag{6}$$

- **Interpretation.** The paper compares the regression R^2before and after the rebate ban.
- **Meaning.** If the ban improves transparency, observable fundamentals should explain a larger share of issuance pricing.

3.7 Conceptual framework

- **Issuer.** Because maturities are short, issuers come back often. The issuance yield is a public benchmark for future debt financing and even bank loans.
- **Underwriter.** Since secondary-market stabilization is not required, the underwriter competes by improving the issuance price rather than by supporting the bond after issuance.
- **Investor.** Secondary-market trading is costly enough that buying in the primary market can still be attractive even when issuance overpricing exists.
- **Regulator.** The rebate ban targets an opaque practice that can distort auction outcomes and lower pricing quality.

4 Identification and instruments (what makes the estimates credible)

4.1 No classical instrument; credibility comes from design

- The baseline result is a near-mechanical comparison of issuance prices to first trading prices. Because over 97% of bonds trade on the first trading day, the comparison window is extremely short.
- The result survives two different overpricing measures and becomes even larger at the 15-day horizon, so it is not a fragile first-trade anomaly.
- Overpricing appears across ratings, maturities, issuer sizes, issuance histories, issuer types, and underwriter types.

4.2 Why the underwriter-incentive story is plausible

- Firms issue repeatedly, so current pricing performance affects future mandates.
- The issuance yield is more salient and more public than thin secondary-market prices, which gives issuers a reason to reward underwriters for high issuance prices.
- Table 4 shows that underperforming underwriters are more likely to be replaced in the next issue.

4.3 The October 2017 rebate ban as quasi-experimental variation

- The ban is exogenous to any single issue but changes underwriters' ability to secretly subsidize auction participation.
- The cross-section gives sharp predictions: a larger drop for central SOE issuers and a smaller drop for Big Four underwriters.
- The paper also uses preregistered sequential issuances spanning the ban, which helps hold issuer composition fixed.

4.4 Ruling out alternative stories for underwriter purchases

- If underwriters self-purchased because they had superior information about undervaluation, self-purchased issues should perform better, not worse.
- If they were merely providing fair-value support, issues they buy should not be systematically more overpriced.
- Instead, issues both underwritten and bought by the same institution are the most overpriced, which supports deliberate overbidding.

5 Tables and figures

5.1 Figure 1: Issuance of debt instruments by category

Read:

- The interbank market expands rapidly from about 1 trillion RMB in 2009 to about 6.6 trillion RMB in 2019.
- The growth is dominated by CP and MTN, not long-term corporate bonds.
- This figure motivates the paper's emphasis on short maturities and repeated issuance.

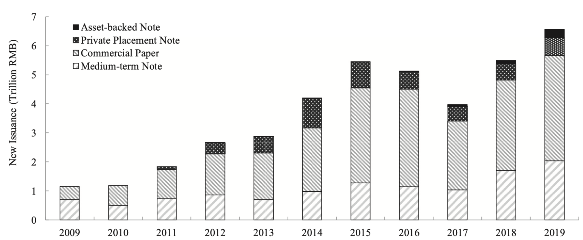

Fig. 1. Issuance of debt instruments by category. This figure plots the issuance amount of debt-financing instruments of non-financial enterprises in the interbank market by category from 2009 to 2019.

Table 1

Summary statistics of debt-security issuance. This table reports summary statistics of the issuance of non-financial corporate debt securities in the interbank market from 2015 to 2019. Panel A reports the number of issuances, issuing companies, and the total issuance amount for each year. Panels B and C report the summary statistics of security and issuer characteristics, respectively. Trading Volume is for the month right after issuance. The subscription ratio is calculated by dividing the total subscription by the issuance amount. The dummy variable First Issue Dummy equals 1, if the security is the issuer's first issuance in the interbank market, and 0 otherwise. Recent Issuance Dummy is another dummy variable and equals 1 if the issuer has issued security in the previous year, and 0 otherwise. We convert letter credit ratings into numerical values, specifically, AAA to 5, AA+ to 4, AA to 3, AA− to 2, and *A*+ to 1. ROA is defined as net income divided by total assets. Sale is the issuer's annual sales. Panel D summarizes the share of issuances directly acquired by underwriters. The number of observations, the mean, the standard deviation, the 25th percentile, the median, and the 75th percentile are reported in Panels B–D.

Panel A: Issuances across years						
	2015	2016	2017	2018	2019	Total
No. of Issues	3,379	3,441	2,880	4,087	4,442	18,229
issued by the Big Four banks	1,258	1,400	957	1,353	1,431	6,399
No. of Companies	1,304	1,238	1,016	1,195	1,354	2,558
Issue Amount (¥bil)	4,457	4,302	3,197	4,488	4,626	21,069
Panel B: Debt-security characteristics						
	N	Mean	SD	P25	P50	P75
Coupon rate (%)	18,229	4.54	1.23	3.55	4.44	5.34
Maturity (year)	18,229	1.74	1.71	0.74	0.76	3.01
Issue Amount (¥mil)	18,229	1,156	1,205	500	1,000	1,500
Trading Volume (¥mil)	18,229	1,350	1,730	440	840	1,609
Subscription Ratio	17,416	1.74	0.88	1.10	1.49	2.08
First Issue Dummy	18,229	0.07	0.26	0.00	0.00	0.00
Recent Issuance Dummy	18,229	0.84	0.36	1.00	1.00	1.00
Rating	18,229	4.18	0.83	4.00	4.00	5.00
Panel C: Issuer characteristics						
	N	Mean	SD	P25	P50	P75
Leverage	18,222	0.65	0.13	0.57	0.66	0.74
ROA	18,219	0.02	0.02	0.01	0.02	0.03
Asset (¥mil)	18,222	163,611	434,672	25,090	55,627	153,839
Sale (¥mil)	18,148	59,335	163,015	4,269	15,051	54,261
Cash (¥mil)	18,148	13,781	38,005	2,266	5,491	14,391
Panel D: Issuances purchase by underwriters						
	N	Mean	SD	P25	P50	P75
Underwriter Share	16,384	0.35	0.31	0.07	0.29	0.56

5.2 Table 1: Summary statistics

Read:

- The sample is economically huge: 18,229 issues and roughly 21.1 trillion RMB of issuance after the trading filter.
- Issuers are large and highly rated; the average maturity is only 1.74 years.
- Underwriters buy about 35% of the securities they underwrite, making self-purchases central to the mechanism analysis.

5.3 Table 2: Baseline issuance overpricing

Read:

- The headline spread-change estimate is **4.90 bps**; the headline excess-return estimate is **-7.67 bps**.
- At 15 days, the average spread change rises to **7.93 bps**, so there is no quick reversal.
- Overpricing falls from **7.44** to **2.41 bps** after the rebate ban.

Table 2

Issuance overpricing. This table reports the summary statistics of the spread change and the excess return after issuance. Panel A reports the summary statistics of ΔSpread, which is the spread difference between the issuance and the first trading day after issuance, $\Delta\text{Spread}_{15\ days}$, which is the spread difference between the issuance and the 15th calendar day since issuance, and the difference between ΔSpread and $\Delta\text{Spread}_{15\ days}$. The spread is calculated as the corporate debt yield minus the corresponding Chinese Treasury Yield Index of similar maturity. Panel B reports the summary statistics of the first-trade excess return, the excess return over 15 calendar days after the issuance, and the difference between the $\text{Excess return}_{15\ days}$ and the Excess return. If the security is not traded on the 15th calendar day, we use the spread or return of the closest trading day within a five-day window centered on the 15th calendar day. We can only calculate the ΔSpread_{15} and $\text{Excess return}_{15\ days}$ for 5,464 issuances due to infrequent trading. The number of observations, the mean, the standard deviation, the *t*-statistic clustered by issuance date, the skewness, the kurtosis, the 5th percentile, the 25th percentile, the median, the 75th percentile, and the 95th percentile are reported. Both spread change and excess return are in basis points (bps). Our sample is from 2015 to 2019, and the rebate ban became effective on October 1, 2017.

Panel A: Spread change (bps)											
Full sample	N	Mean	SD	*t*-Stat.	Skew.	Kurt.	P5	P25	P50	P75	P95
ΔSpread	18,229	4.90	12.30	26.46	3.85	37.97	−6.87	−0.82	2.55	8.08	23.35
$\Delta\text{Spread}_{15\ days}$	5,464	7.93	39.41	12.18	9.17	268.99	−35.00	−7.25	4.71	17.94	58.44
$\Delta\text{Spread}_{15\ days}$ -ΔSpread	5,464	1.96	38.15	3.22	9.97	308.23	−39.62	−11.23	−0.17	11.60	46.50
Before rebate ban											
ΔSpread	9,026	7.44	11.00	36.74	2.74	42.86	−4.82	1.96	6.57	10.70	26.28
$\Delta\text{Spread}_{15\ days}$	2,984	10.53	38.03	11.57	1.66	14.57	−34.81	−5.63	7.37	21.34	66.56
$\Delta\text{Spread}_{15\ days}$ -ΔSpread	2,984	2.69	37.62	3.07	1.56	14.82	−45.48	−12.26	−0.50	12.95	55.99
After rebate ban											
ΔSpread	9,203	2.41	12.97	9.41	5.02	41.16	−7.68	−2.10	0.36	3.23	15.99
$\Delta\text{Spread}_{15\ days}$	2,480	4.81	40.79	5.45	16.63	507.77	−35.28	−8.54	2.17	13.69	50.12
$\Delta\text{Spread}_{15\ days}$ -ΔSpread	2,480	1.07	38.77	1.30	19.23	622.97	−35.26	−9.92	0.18	10.11	36.73
Panel B: Excess return (bps)											
Full sample	N	Mean	SD	*t*-Stat.	Skew.	Kurt.	P5	P25	P50	P75	P95
Excess return	18,229	−7.67	10.50	−42.72	−2.52	62.38	−22.43	−11.60	−6.20	−2.46	3.52
$\text{Excess return}_{15\ days}$	5,464	−12.46	44.38	−13.93	−0.84	27.29	−74.60	−29.40	−12.00	6.37	47.31
$\text{Excess return}_{15\ days}$ - Excess return	5,464	−4.08	42.60	−5.15	−0.86	33.29	−60.21	−19.48	−3.39	11.88	52.80
Before rebate ban											
Excess return	9,026	−10.30	11.14	−40.87	−2.40	54.20	−25.44	−14.90	−9.65	−4.91	3.04
$\text{Excess return}_{15\ days}$	2,984	−16.92	45.85	−13.11	−0.05	12.61	−85.44	−35.63	−16.09	4.69	43.16
$\text{Excess return}_{15\ days}$ - Excess return	2,984	−6.39	44.11	−5.61	0.13	15.54	−69.75	−22.77	−4.71	12.77	49.34
After rebate ban											
Excess return	9,203	−5.08	9.12	−26.79	−2.92	95.86	−16.48	−7.14	−4.00	−1.44	3.83
$\text{Excess return}_{15\ days}$	2,480	−7.11	41.93	−6.30	−2.06	54.60	−57.02	−23.39	−8.78	7.87	52.08
$\text{Excess return}_{15\ days}$ - Excess return	2,480	−1.30	40.55	−1.24	−2.37	63.86	−48.49	−15.71	−2.02	10.69	56.41

Table 3

Overpricing across security and issuer characteristics. This table reports the first trading day spread change in basis points (bps), ΔSpread, across different debt ratings, maturities, issuers' total asset, and issuing history, as well as issuer and underwriter types in the periods before and after the rebate ban. The number of observations, the mean, and the *t*-statistics clustered by issuance date are presented. The sample is from 2015 to 2019, and the rebate ban became effective on October 1, 2017.

		Full sample			Before rebate ban			After rebate ban		
Panel A: Sort by rating (bps)		N	Mean	*t*-Stat.	N	Mean	*t*-Stat.	N	Mean	*t*-Stat.
AAA		8,038	6.53	25.19	3,433	9.33	33.12	4,605	4.44	12.02
AA+		5,706	3.23	15.98	2,665	6.38	26.82	3,041	0.47	1.90
AA		4,275	4.03	19.23	2,724	6.22	25.09	1,551	0.19	0.78
AA- and *A+*		210	5.84	7.87	204	5.88	7.72	6	4.41	2.22
Panel B: Sort by rating and maturity (bps)		N	Mean	*t*-Stat.	N	Mean	*t*-Stat.	N	Mean	*t*-Stat.
	Maturity									
AAA	<1 year	4,905	9.18	24.82	2,248	11.99	32.88	2,657	6.80	11.75
	1–2 year	734	4.76	10.87	394	7.71	12.49	340	1.34	2.55
	≥2 year	2,399	1.65	8.24	791	2.57	7.34	1,608	1.20	4.96
AA+	<1 year	3,001	4.06	16.07	1,306	8.73	30.20	1,695	0.47	1.69
	1–2 year	1,005	4.23	10.13	621	6.63	14.68	384	0.34	0.48
	≥2 year	1,700	1.17	5.09	738	2.00	6.10	962	0.53	1.71
AA, AA-, and *A+*	<1 year	1,658	5.04	16.23	979	8.45	22.49	679	0.12	0.38
	1–2 year	1,289	6.06	16.20	1,074	7.21	17.44	215	0.34	0.56
	≥2 year	1,538	1.49	7.85	875	2.44	9.13	663	0.25	1.02
Panel C: Sort by rating and total assets (bps)		N	Mean	*t*-Stat.	N	Mean	*t*-Stat.	N	Mean	*t*-Stat.
	Total Assets									
AAA	Larger	4,026	7.69	22.17	1,718	10.17	26.35	2,314	5.92	11.68
	Smaller	4,012	5.36	19.18	1,715	8.48	24.63	2,291	2.94	7.86
AA+	Larger	2,853	3.25	13.35	1,338	6.59	21.43	1,521	0.73	2.43
	Smaller	2,853	3.21	14.28	1,327	6.16	21.67	1,520	0.21	0.79
AA, AA-, and *A+*	Larger	2,244	3.86	16.01	1,465	6.38	21.39	779	0.11	0.40
	Smaller	2,241	4.37	17.68	1,463	6.02	19.70	778	0.31	1.10
Panel D: Sort by issuance history (bps)		N	Mean	*t*-Stat.	N	Mean	*t*-Stat.	N	Mean	*t*-Stat.
First-time issuance		1,305	2.82	11.53	762	4.40	13.63	543	0.61	1.94
Seasoned offering		16,924	5.06	26.45	8,264	7.72	36.75	8,660	2.52	9.60
Panel E: Sort by issuer type (bps)		N	Mean	*t*-Stat.	N	Mean	*t*-Stat.	N	Mean	*t*-Stat.
Central SOE		1,635	10.31	20.15	923	12.24	22.99	712	7.81	8.39
Other		16,594	4.37	23.92	8,103	6.89	34.11	8,491	1.96	7.85
Panel F: Sort by underwriter type (bps)		N	Mean	*t*-Stat.	N	Mean	*t*-Stat.	N	Mean	*t*-Stat.
Big Four banks		6,399	4.71	22.20	3,415	7.53	30.87	2,984	1.49	5.38
Other		11,830	5.00	24.28	5,611	7.39	31.70	6,219	2.86	9.67

5.4 Table 3: Heterogeneity across ratings, maturities, issuers, and underwriters

Read:

- Overpricing is broad-based: AAA issues show **6.53 bps**, AA+ issues **3.23 bps**, and AA issues **4.03 bps**.
- It is larger for shorter maturities, especially AAA debt below one year (**9.18 bps**).
- Central-SOE issues are much more overpriced (**10.31 bps**) than other issues (**4.37 bps**).
- After the ban, overpricing remains strongest in AAA issues, consistent with safer bonds being the easiest place to overbid.

5.5 Table 4: Underwriter switching and repeat business

Read:

- Underwriters that price an issue worse than comparable issues are more likely to be replaced in the next issuance.
- The paper interprets column (1) as implying roughly an **8.3%** increase in switching probability when the issue underperforms.
- The relationship weakens after the rebate ban but does not vanish.

Table 4
Logit regression of underwriter switching. This table reports the logit regressions of an issuer's underwriter change on the underwriter's performance in the issuer's last debt-security issuance. The dependent variable, $Switch_{j,n+1}$, equals 1 if issuer j changes the underwriters of its $n+1$th issuance as compared to its n^{th} issuance, and 0 otherwise. Performance is measured by an indicator variable, $Underperformed_{j,n}$, which equals 1 if the spread of issuer j's n^{th} issuance is greater than the corresponding benchmark spread. $Underwriter\ Share_{j,n}$ is the share purchased by the underwriter in issuer j's n^{th} issuance. Heteroskedasticity-consistent z-statistics clustered by issuance date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent: $Switch_{j,n+1}$	Full sample (1)	Full sample (2)	Before ban (3)	After ban (4)	Full sample (5)
$Underperformed_{j,n}$	0.281***	0.212***	0.288***	0.105**	0.202***
	(8.77)	(5.87)	(5.73)	(2.01)	(5.26)
$Underwriter\ Share_{j,n}$					−0.222***
					(−3.46)
Ln(Issue Amount)		−0.017	0.067	−0.086	−0.018
		(−0.38)	(1.12)	(−1.19)	(−0.37)
Subscription Ratio		−0.022	−0.009	−0.040	−0.038*
		(−1.02)	(−0.29)	(−1.38)	(−1.67)
Maturity		−0.037***	−0.020	−0.054***	−0.039***
		(−3.18)	(−1.26)	(−3.12)	(−3.14)
Ln(Trading Volume)		−0.034	0.026	−0.065	−0.052*
		(−1.18)	(0.69)	(−1.45)	(−1.70)
First Issue Dummy		−0.140	−0.011	−0.344**	−0.141
		(−1.40)	(−0.09)	(−2.27)	(−1.32)
Recent Issuance Dummy		0.785***	0.709***	0.878***	0.791***
		(11.08)	(7.21)	(8.50)	(10.38)
$Dummy_{AAA}$		0.789***	0.612**	0.563***	0.939**
		(3.11)	(2.35)	(5.03)	(2.18)
$Dummy_{AA+}$		0.529**	0.446*	0.183**	0.660
		(2.13)	(1.78)	(2.17)	(1.55)
$Dummy_{AA}$		0.355	0.300		0.523
		(1.45)	(1.22)		(1.24)
Leverage		1.002***	1.243***	0.824***	1.066***
		(5.63)	(5.20)	(3.13)	(5.71)
ROA		−1.990**	−0.965	−3.495**	−1.950**
		(−2.14)	(−0.85)	(−2.14)	(−1.96)
Ln(Asset)		0.240***	0.154***	0.270***	0.272***
		(6.51)	(3.04)	(4.78)	(6.92)
Ln(Sales)		0.033**	0.038	0.052**	0.042**
		(2.12)	(1.58)	(2.40)	(2.54)
Ln(Cash)		−0.139***	−0.112***	−0.188***	−0.166***
		(−5.21)	(−3.02)	(−4.83)	(−5.90)
Constant	0.390***	−2.815***	−3.290***	−1.698***	−2.962***
	(17.77)	(−7.83)	(−7.90)	(−3.82)	(−5.80)
Observations	16,920	15,958	8,131	7,827	14,311
Pseudo R-squared	0.00331	0.0575	0.0537	0.0621	0.0599

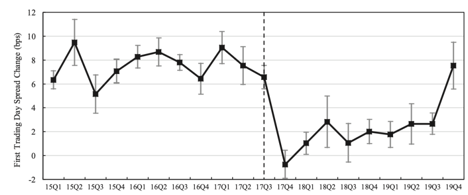

Fig. 2. Issuance overpricing over time. This figure plots the quarterly issuance overpricing along with their 95% confidence intervals from 2015 to 2019. The standard error is clustered by issuance date.

5.6 Figure 2: Issuance overpricing over time

Read:

- Quarterly overpricing is clearly positive before late 2017 and drops sharply around the rebate ban.
- Overpricing remains positive afterward, which visually foreshadows the paper's two-channel interpretation: rebates weaken, but self-purchases remain.

5.7 Tables 5 and 6: Difference-in-differences around the rebate ban

Read:

- **Table 5:** in the matched sample with controls, the interaction for central SOE issues is about **-7.86 bps**. These issuers experience a much larger decline in overpricing after the ban.
- **Table 6:** in the matched sample with controls, the interaction for Big Four underwriters is about **+2.32 bps**. The drop is smaller where underwriter market power is already high.
- Together, the two tables strongly support a competition-and-rebates channel.

5.8 Tables 7 and 8: Underwriter self-purchases and overbidding

Read:

- Underwriters buy more in overpriced issues (purchase share **0.37**) than in other issues (**0.29**).

Table 5

Effect of the rebate ban on overpricing: variation across issuers. This table reports results of the difference-in-differences analysis of how the rebate ban affected issuance overpricing. The sample includes all MTN and CP issued by nonfinancial firms in China's interbank market from April 1, 2017, to March 31, 2018, a 12-month window around the rebate ban on October 1, 2017. Treat equals 1 if the issuance is by a central SOE, and 0 otherwise. Post equals 1 in the months following the rebate ban. Columns (1) and (2) use the full sample. Columns (3) and (4) use the matched sample, which includes only sequential issuances before and after the rebate ban. Heteroskedasticity-consistent t-statistics clustered by issuance date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Full sample		Matched sample	
Dependent: ΔSpread (bps)	(1)	(2)	(3)	(4)
Treat	9.709***	6.441***	9.772***	6.545***
	(6.37)	(4.50)	(5.06)	(3.54)
Post	−6.043***	−6.273***	−7.043***	−7.139***
	(−10.70)	(−11.17)	(−9.16)	(−9.48)
Treat × Post	−7.225***	−6.182***	−8.407***	−7.861***
	(−3.62)	(−3.22)	(−3.65)	(−3.71)
Ln(Issue Amount)		0.430		1.839*
		(0.67)		(1.87)
Subscription Ratio		−0.741***		−0.950**
		(−3.50)		(−2.29)
Maturity		−1.206***		−1.636***
		(−8.81)		(−7.06)
Ln(Trading Volume)		−0.071		−0.282
		(−0.19)		(−0.47)
First Issue Dummy		0.624		2.420
		(0.96)		(0.98)
Recent Issuance Dummy		0.113		0.197
		(0.22)		(0.10)
$Dummy_{AAA}$		0.068		−1.017
		(0.10)		(−0.88)
$Dummy_{AA+}$		−0.335		−0.912
		(−0.84)		(−1.44)
Leverage		−2.215		−4.098
		(−1.35)		(−1.51)
ROA		12.641**		11.949
		(2.10)		(1.06)
Ln(Asset)		1.537***		1.861**
		(3.50)		(2.58)
Ln(Sales)		−0.067		−0.180
		(−0.47)		(−0.84)
Ln(Cash)		−1.078***		−1.356***
		(−3.54)		(−3.01)
Constant	6.203***	1.379	7.659***	−3.606
	(14.65)	(0.42)	(12.11)	(−0.64)
Observations	3,252	3,164	1,481	1,445
R-squared	0.153	0.210	0.182	0.246

Table 6

Effect of the rebate ban on overpricing: variation across underwriters. This table reports results of the difference-in-differences analysis of how the rebate ban affected issuance overpricing. The sample includes all MTN and CP issued by nonfinancial firms in China's interbank market from April 1, 2017, to March 31, 2018, a 12-month window around the rebate ban on October 1, 2017. Treat equals 1 if the issuance is underwritten by one of the Big Four banks in China, and 0 otherwise. Post equals 1 in the months following the rebate ban. Columns (1) and (2) use the full sample. Columns (3) and (4) use the matched sample, which includes only sequential issuances before and after the rebate ban. Heteroskedasticity-consistent t-statistics clustered by issuance date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Full sample		Matched sample	
Dependent: ΔSpread (bps)	(1)	(2)	(3)	(4)
Treat	−0.791	−1.536***	−2.459***	−2.698***
	(−1.39)	(−2.89)	(−2.96)	(−3.35)
Post	−7.187***	−7.363***	−9.147***	−8.842***
	(−11.51)	(−12.04)	(−10.00)	(−10.20)
Treat × Post	1.362*	1.616**	2.712**	2.316**
	(1.89)	(2.37)	(2.51)	(2.34)
Issuance Controls	No	Yes	No	Yes
Firm Controls	No	Yes	No	Yes
Constant	7.239***	−4.492	9.699***	−10.431*
	(15.17)	(−1.40)	(12.35)	(−1.87)
Observations	3,252	3,164	1,481	1,445
R-squared	0.119	0.200	0.149	0.232

Table 7

Underwriter purchases and overpricing. Panel A reports summary statistics of the share purchase by the underwriter, *Underwriter Share*, across issuances with and without overpricing, as well as across different ratings, issuer and underwriter types, and sample periods. Number of observations, the mean, the standard deviation, the 25th percentile, the median, and the 75th percentile are presented. Our sample is from 2015 to 2019, and the rebate ban became effective on October 1, 2017. Panel B reports the average overpricing (in basis points) of issuances acquired by qualified investors (column 1), acquired by licensed underwriters but underwritten by others (column 2), and acquired and underwritten by the same licensed underwriters (column 3). We first calculate both the equal-weighted average spread change and the value-weighted average spread change (using purchase amount as the weight) for each institution and then take the average across the institutions in each category. Panel B also reports t-statistics for the differences between (1) and (3) and between (2) and (3), with *, **, and *** indicating statistical significance at the 10%, 5% and 1% levels, respectively.

Panel A. Summary statistics of underwriter purchase

Underwriter purchase by overpricing	N	Mean	SD	P25	P50	P75
Overpriced issuances	11,058	0.37	0.31	0.10	0.32	0.60
Other	5,326	0.29	0.29	0.00	0.20	0.46
Underwriter purchase by rating						
AAA	7,321	0.38	0.32	0.10	0.31	0.60
AA+	5,239	0.29	0.28	0.00	0.22	0.50
AA	3,720	0.35	0.31	0.06	0.30	0.56
AA- and *A+*	104	0.60	0.32	0.30	0.68	0.86
Underwriter purchase by issuer type						
Central SOE	1,405	0.49	0.31	0.23	0.47	0.74
Other	14,979	0.33	0.30	0.05	0.27	0.54
Underwriter purchase by underwriter type						
Big Four banks	5,594	0.36	0.30	0.10	0.30	0.56
Other	10,790	0.34	0.31	0.05	0.27	0.56
Underwriter purchase by rebate ban						
Before rebate ban	7,191	0.44	0.30	0.20	0.44	0.68
After rebate ban	9,193	0.27	0.29	0.00	0.20	0.41

Panel B. Overpricing of issuances acquired by qualified investors and licensed underwriters

	Overpricing of issuances acquired by qualified investors (1)	Overpricing of issuances acquired by licensed underwriters but underwritten by others (2)	Overpricing of issuances acquired and underwritten by the same licensed underwriters (3)	Difference (3)-(1)	Difference (3)-(2)
Equal-weighted portfolio average					
Full sample	1.54	2.19	3.85	2.32***	1.67***
Before rebate ban	4.95	5.39	7.35	2.40**	1.96**
After rebate ban	0.66	1.01	2.19	1.52**	1.18**
Value-weighted portfolio average					
Full sample	1.57	2.95	6.40	4.83***	3.45***
Before rebate ban	5.49	5.80	8.15	2.65**	2.35**
After rebate ban	0.50	1.50	5.61	5.11***	4.11***

Table 8

Regressions of overpricing on underwriter purchases. This table reports regressions of issuance overpricing on the share purchase by the underwriter. The dependent variable is the overpricing measure, ΔSpread. The independent variable *Underwriter Share* is the share purchased by the underwriter. Columns (1) and (2) report regression results for the full sample. Columns (3) and (4) report regression results for issuances before and after the rebate ban, respectively. Heteroskedasticity-consistent t-statistics clustered by issuance date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent: ΔSpread (bps)	Full sample (1)	Full sample (2)	Before ban (3)	After ban (4)
Underwriter Share	10.494***	9.118***	1.802**	14.943***
	(17.32)	(16.43)	(2.06)	(12.71)
Issuance Controls	No	Yes	Yes	Yes
Firm Controls	No	Yes	Yes	Yes
Constant	1.004***	−8.392***	−1.458	−3.128
	(4.10)	(−5.13)	(−0.78)	(−0.91)
Observations	16,384	15,465	7,091	8,374
R-squared	0.069	0.120	0.118	0.144

Table 9
Quality of issuance price. This table reports regressions of issuance yield spread on issuance and issuer characteristics. The dependent variable is $\text{Spread}_{issuance}$, measured as the coupon rate minus Treasury yield with similar maturity. Columns (1)–(4) report the regression results for all issuances in each of the four years around the rebate ban, respectively. Heteroskedasticity-consistent t-statistics clustered by issuance date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent: $\text{Spread}_{issuance}$ (%)	Two years before rebate ban (1)	One year before rebate ban (2)	One year after rebate ban (3)	Two years after rebate ban (4)
Ln(Issue Amount)	−0.001	−0.055***	−0.156***	−0.164***
	(−0.03)	(−2.78)	(−5.95)	(−6.24)
Maturity	0.095***	0.103***	0.087***	0.108***
	(12.81)	(10.53)	(9.95)	(12.58)
First Issue Dummy	0.067	−0.158***	0.018	0.051
	(1.11)	(−2.74)	(0.30)	(0.61)
Recent Issuance Dummy	0.044	−0.038	0.136***	0.041
	(0.83)	(−0.83)	(2.86)	(0.70)
Dummy_{AAA}	−2.930***	−1.507***	−1.897***	−1.835**
	(−20.11)	(−8.89)	(−6.03)	(−2.43)
Dummy_{AA+}	−2.391***	−0.901***	−0.990***	−0.783
	(−17.07)	(−5.37)	(−3.20)	(−1.03)
Dummy_{AA}	−1.728***	−0.383**	−0.347	0.022
	(−12.55)	(−2.30)	(−1.13)	(0.03)
Leverage	0.472***	0.796***	0.603***	1.032***
	(4.89)	(6.92)	(4.84)	(7.83)
ROA	−4.691***	1.409*	0.174	−0.370
	(−6.56)	(1.90)	(0.23)	(−0.58)
Ln(Asset)	−0.163***	−0.079***	−0.105***	−0.124***
	(−6.99)	(−3.22)	(−4.04)	(−4.33)
Ln(Sales)	0.173***	0.082***	0.062***	0.100***
	(14.80)	(6.68)	(5.46)	(9.53)
Ln(Cash)	0.066***	0.117***	0.150***	0.121***
	(4.81)	(6.81)	(8.59)	(5.86)
Constant	2.884***	1.573***	3.136***	2.465***
	(15.50)	(6.33)	(9.55)	(3.16)
Observations	3,610	2,942	3,562	4,517
R-squared	0.339	0.348	0.436	0.392

- Issues both acquired and underwritten by the same licensed underwriters are the most overpriced: equal-weighted overpricing is **3.85 bps** versus **1.54 bps** for qualified investors; value-weighted it is **6.40 bps** versus **1.57 bps**.
- In Table 8, the coefficient on *Underwriter Share* is positive and strong: about **9.12** with controls in the full sample, **1.80** before the ban, and **14.94** after the ban.
- This is powerful evidence against informed bargain-hunting and in favor of deliberate overbidding.

5.9 Table 9: Quality of issuance price

Read:

- The $R^2$of issuance-spread regressions rises from about **0.339** and **0.348** before the rebate ban to about **0.436** and **0.392** after it.
- The interpretation is that the ban not only reduced overpricing but also made issuance prices more tightly linked to fundamentals.

6 Short summary

- **Conclusion.** China's interbank corporate debt market exhibits systematic issuance overpricing.
- **Intuition.** Underwriters compete for repeat business by delivering high issuance prices to issuers that care about current benchmark financing rates.
- **Empirical lesson.** Institutional details of auction design, investor base, and underwriting incentives can reverse the usual sign of new-issue pricing.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- New issues of Chinese corporate debt are overpriced at issuance, not underpriced.
- The magnitude is meaningful: 4.9 bps in spread terms and -7.67 bps in return terms.

- The rebate ban substantially reduces, but does not eliminate, overpricing.
- Mechanism evidence points to underwriter competition, rebates, and self-purchases/overbidding.
- The post-ban increase in pricing R^2 suggests the regulation improved pricing quality.

7.2 Economic intuition

- Issuance yields matter because they are public, salient, and tied to future financing conditions.
- Secondary-market illiquidity weakens market discipline after the issue.
- Banks' dual role as underwriters and investors makes it feasible to sustain overpricing even when explicit rebates are banned.

7.3 Takeaway

This paper shows that primary-market pricing is not a universal object. In China's interbank debt market, underwriters are rewarded for making issuers look cheap to fund, not for leaving rents to investors. That flips the familiar underpricing result into systematic overpricing, and the paper traces the flip directly to the market's institutional design.